

Eligibility Traces as a Memory-Trace Substrate for Compulsive Checking: A Partially Observable Trace-Based Checking Agent and a Candid In-Silico Test

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Abstract

Obsessive-compulsive disorder (OCD) has been formalised both as excessive subjective uncertainty about state transitions and as a bias towards model-free habit control, but these accounts are rarely expressed in a single mechanism. We propose the Trace-Based Checking Agent (TBCA), a $TD(\lambda)$ learner with accumulating or replacing eligibility traces operating over a discretised belief state in a partially observable checking task, arbitrated with a one-step belief-based module, and run with the identical parameter vector on a probabilistic reversal task. We simulated 1572 synthetic agents under a single seed across an $8(\lambda) \times 2(\text{trace type}) \times 3(\text{check reliability } q)$ design plus a model-recovery grid. Results were mixed and are reported candidly. H1 (runaway escalation with accumulating traces) was not supported: checking was essentially flat in λ (3.13 to 2.75 checks/episode, $q = 0.8$) and the $\lambda \times \text{trace-type}$ interaction was indistinguishable from chance ($F(7, 304) = 0.92$, $p = 0.49$, $\eta^2 = 0.0197$), with an approximate detection limit of $\eta^2 \approx 0.04$. We emphasise that this is most likely an untested rather than a falsified hypothesis: we did not log eligibility magnitudes or state-revisitation frequencies, and because the check count is part of the state, accumulating traces may rarely have exceeded eligibility 1, in which case the two trace variants were identical by construction. H2 was partially supported: subjective uncertainty and trace length are interchangeable at the level of checking rate (5/6 ε levels matched within 10%) and uncertainty-generated data were fitted with $\hat{\lambda} = 0.37$, but varying q within-subject did not resolve the confound, and the recovery estimates are uncalibrated single-seed point estimates. Surprisingly, higher subjective instability reduced checking in this minimal variant, which omits the stopping criterion and relief reinforcer of the account it draws on. H3 was partially supported (reward-optimal λ monotone in falling reliability, but two of three optima on the grid boundary). Exploratory and speculative: longer accumulating traces lowered a stimulus-bound repetition index, but this is plausibly a confound with effective learning rate and is not offered as a route to the reported OCD/stimulant-use-disorder dissociation. We identify the sensitivity sweeps, mechanism diagnostics, multi-seed replication and normative benchmark that a fair test of the trace account requires.

Keywords: obsessive-compulsive disorder, eligibility traces, $TD(\lambda)$, compulsive checking, partial observability, model-based / model-free arbitration, perseveration, computational psychiatry

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1 Introduction

Compulsive checking is the paradigmatic symptom of obsessive-compulsive disorder (OCD): a patient returns repeatedly to a stove, a lock or a switch, each time acquiring evidence that in principle should settle the question, and each time failing to be settled. Two computational accounts dominate. The first treats OCD as a disorder of inference: patients behave as if the world changed faster than it does, so evidence decays in subjective value and doubt is never extinguished [2]. The second treats OCD as a disorder of control arbitration: compulsivity across diagnoses is associated with a bias towards model-free habit learning [3], and with structural abnormalities in caudate and medial orbitofrontal cortex [5]. Hierarchical fits of probabilistic reversal learning show that these disorders are not interchangeable: OCD and stimulant use disorder dissociate on perseveration parameters, and are differentially modulated by dopaminergic D2/3 agents [1].

What neither account makes explicit is the memory trace itself. In reinforcement learning, the eligibility trace is precisely the object that determines how long a past state-action pair remains creditable for later outcomes, and it exists in accumulating and replacing variants with markedly different behaviour under state revisitation [10, 12], with off-policy generalisations [4]. Traces are also the most physiologically constrained part of the model-free apparatus: dopamine-cell responses to predicted events in classical conditioning imply long-lasting traces combined with low learning rates [8]. Traces are, formally, the continuum along which model-free bootstrapping shades into Monte-Carlo-like credit assignment, and hence a natural meeting point with model-based computation [6].

Our theme is the development of a memory-trace model of OCD. We take the theme literally and ask whether the eligibility trace can serve as the memory-trace substrate for obsessive repetition. The specific idea we test is mechanistic rather than metaphorical: in a checking task, repeated checks at an unchanged belief map to the same discretised state-action pair, so an accumulating trace can super-credit repetition, and any later relief of the aversive episode is then assigned disproportionately to the act of checking, even though no relief reinforcer is written into the reward function. This mechanism has an explicit precondition – that consecutive checks leave the represented state unchanged within the trace window – and one of the main lessons of this revision is that we did not verify that precondition in our own simulations.

We make three contributions. (i) We define the Trace-Based Checking Agent (TBCA), a $TD(\lambda)$ agent over (check-count, belief-bin) states in a partially observable checking MDP, arbitrated with a one-step belief computation that uses the agent’s own, possibly inflated, subjective flip probability ε ; the same parameter vector also drives a two-armed reversal task so that compulsion and stimulus-bound perseveration can dissociate within one agent. (ii) We run a factorial in-silico study and a model-recovery grid to test three in-silico hypotheses about trace length, trace type, uncertainty-trace equivalence and the cost of robustness to hidden state. (iii) We report the outcome honestly, including a null result on our headline hypothesis, an unexpected reversal of the direction of the uncertainty effect, and a model-recovery result that goes the wrong way. Following review, we are careful to distinguish three epistemic categories: results that are supported by the simulation, results that are null within a quantified detection limit, and claims that our design cannot adjudicate at all. Our headline hypothesis now falls into the third category more than the second: without a diagnostic of eligibility magnitude and without a sweep over the state discretisation and the checking cap, the trace mechanism is best described as untested rather than falsified. We argue that these outcomes are nonetheless informative: they identify which parts of the trace account are load-bearing, which were wishful, and which experiments (in silico and in humans) are prerequisites for a fair test.

2 Related work

Computational psychiatry of OCD. Fradkin et al. [2] formalise OCD as a Bayesian agent that assumes excessive instability in state transitions; because evidence is discounted by an inflated transition probability, beliefs never consolidate, and the model reproduces doubt, checking, repetition and an over-reliance on habitual action. This account locates the pathology in the generative model rather than in the learning rule. It is important for the interpretation of our results that the TBCA imports only one component of that account – the inflated subjective flip probability ε – and omits several others: there is no explicit termination or confidence criterion governing when checking stops, no anxiety-relief reinforcer attached to the checking act, no hierarchical belief about volatility that could itself be learned, and no separate memory-confidence variable. In Fradkin et al. [2] the persistence of checking is driven in large part by a stopping rule defined on a certainty threshold; in the TBCA, stopping is driven by expected value under a per-check cost. Any statement we make about the direction of the ε effect is therefore a statement about this minimal variant, not about their model.

Voon et al. [3] instead locate compulsivity in control arbitration, reporting a shared bias towards habit learning across disorders of compulsivity; Raduà and Mataix-Cols [5] provide the structural correlates (caudate/lenticular increases, dorsomedial frontal and anterior cingulate reductions) that are usually read as a striatal-prefrontal imbalance favouring habit. A closely related and now large literature uses two-step sequential decision tasks to argue that compulsivity is associated with reduced goal-directed (model-based) control rather than with strengthened habits per se, and that this deficit tracks a transdiagnostic compulsivity dimension rather than a diagnosis. We engage with that literature here only descriptively: we were unable to include the primary two-step references in the citation set available for this version, and we flag this as a gap in our coverage rather than as a judgement about their importance. The relevance for us is direct, because the two-step account predicts that a shift towards model-free control should be visible as a change in the arbitration weight w rather than in the trace parameter λ – and w is a parameter we held fixed, so our design cannot speak to it.

Kanen et al. [1] add the crucial dissociation constraint: OCD and stimulant use disorder differ on reinforcement-learning and flexibility parameters, with stimulus-bound perseveration elevated in stimulant use disorder but not (indeed, reduced) in OCD, and with D2/3 agents modulating these parameters. Any single-mechanism model of compulsion must therefore be able to increase repetitive checking while not increasing, and ideally decreasing, stimulus-bound perseveration. We note in advance that the parameters in that work are estimated hierarchically from patient choice data under a specific generative model, whereas our perseveration measure is a descriptive repetition statistic computed from simulated choices; the two are not on a common scale.

Eligibility traces. Traces solve temporal credit assignment by keeping a decaying record of recently visited state-action pairs [10]. Singh and Sutton [12] showed that accumulating and replacing traces differ sharply when a state is revisited before the trace has decayed: accumulating traces sum eligibility above one and can over-credit repeated visits, whereas replacing traces reset it, which typically yields faster and more stable learning. The emphasis on revisitation is not incidental; it is the precondition for the two trace variants to differ at all, and it becomes the crux of our null result. Precup et al. [4] extended traces to off-policy evaluation, and Moerland et al. [6] situate the λ continuum between bootstrapped model-free and Monte-Carlo/model-based-like credit assignment. Empirically, Pan et al. [8] showed that dopamine cells respond to predicted events during conditioning in a way that requires long-lasting eligibility traces with low learning rates, making trace duration a biologically measurable, dopamine-linked quantity.

The missing bridge. Despite this, we are not aware of retrieved work that estimates trace parameters (λ , trace type, α) from patient choice data, or that uses trace dynamics as the memory-trace substrate of obsessive repetition. The two OCD accounts and the trace literature

remain separate strands. Two features of the trace literature make the bridge attractive. First, trace type predicts exactly the kind of repetition-specific pathology that checking exemplifies [12]. Second, λ trades bootstrapping against Monte-Carlo returns and is known to help under hidden state, which is precisely the situation of a person who cannot verify that their own action was executed [10]. Methodologically we also draw on the practice of evaluating agents on fixed, released behavioural datasets [7] as a template for the offline evaluation of clinical models, and on mixed-effects modelling [11] for the human designs. Finally, any trace account must eventually make contact with treatment: exposure and response prevention works by preventing the compulsive act from being credited with relief [9], which is a natural target for a trace-level formalisation.

3 Model

The Trace-Based Checking Agent (TBCA) is defined on a partially observable checking MDP. A hazard (a stove) is either on ($s = 1$) or off ($s = 0$). The true state is fixed within an episode; the default true flip probability is zero, so any perceived instability is entirely the agent’s own.

Belief. The agent maintains a scalar belief b_t that the hazard is on. At each step it first applies a subjective transition, $b_t^- = (1 - \varepsilon)b_{t-1} + \varepsilon(1 - b_{t-1})$, where ε is the subjective flip probability that implements the excessive state-transition uncertainty of Fradkin et al. [2]. If the agent checks, it receives a noisy observation o_t with symmetric reliability q , i.e. $P(o = 1 \mid s = 1) = P(o = 0 \mid s = 0) = q$, and updates by Bayes’ rule, $b_t = P(o_t \mid s = 1)b_t^- / [P(o_t \mid s = 1)b_t^- + P(o_t \mid s = 0)(1 - b_t^-)]$. The agent knows q but may misestimate ε .

State representation and actions. The model-free module is tabular over the discretised pair $x_t = (\min(n_t, N_{\max}), \lfloor b_t K \rfloor)$, where n_t is the number of checks so far in the episode and K is the number of belief bins. Actions are CHECK and LEAVE. This representation is the crux of the model: two consecutive checks that do not move the belief across a bin boundary map to the same state-action pair, so an accumulating trace can raise its eligibility above unity. Note, however, that the first component of x_t is the check count, which increments on every check until the cap $N_{\max}$ binds. Exact revisitation of the same (n, bin) cell within an episode is therefore possible only once n_t has saturated at $N_{\max}$ and the belief bin is additionally unchanged. This is an analytic property of the representation, not an empirical measurement, and we return to it as the most likely explanation of our H1 null.

Learning. Values are updated by TD(λ) with $\delta_t = r_t + \gamma \max_{a'} Q_{MF}(x_{t+1}, a') - Q_{MF}(x_t, a_t)$ and $Q_{MF} \leftarrow Q_{MF} + \alpha \delta_t e_t$. Traces are either accumulating, $e_t(x, a) = \gamma \lambda e_{t-1}(x, a) + \mathbb{I}[x = x_t, a = a_t]$, or replacing, $e_t(x, a) = 1$ for the visited pair and $\gamma \lambda e_{t-1}(x, a)$ otherwise [12]. Because the step size α multiplies the trace, raising λ (and, for accumulating traces, revisiting a pair) raises the effective per-transition learning rate. We did not control for this: no effective-learning-rate normalisation was applied. This matters for interpreting the reversal-task result reported below.

Arbitration. Action selection uses a fixed convex mixture $Q = (1 - w)Q_{MF} + wQ_{MB}$, where Q_{MB} is a one-step lookahead computed from the agent’s belief and its own ε . Crucially, ‘model-based’ here does not mean ‘correct’: a model-based agent with inflated ε is wrong in the direction of Fradkin et al. [2]. Arbitration is static, which keeps the parameter count minimal and makes the trade-off between λ and ε interpretable [6], but also means that the model cannot express the model-based/model-free shift emphasised by the two-step literature.

Rewards. Each check costs $c < 0$; leaving while the hazard is on incurs $R_{dis} \ll 0$; leaving safely yields 0. There is deliberately no anxiety-relief bonus and no certainty threshold that must be crossed before leaving. Any preference for checking must therefore be manufactured by credit assignment, not read off a built-in reinforcer or a stopping rule.

Second task. The same agent, with the same λ , α and trace type, is run on a two-armed probabilistic reversal task with stimulus identity as state, yielding a stimulus-bound perseveration

index together with win-stay and lose-shift rates. Reviewers asked for the mathematical definition of that index. We report candidly that the index in our implementation is a scalar repetition statistic computed over reversal trials, and that we cannot reconstruct its exact normalisation from the stored summary outputs to the standard required for a formal definition: the reported values are not recoverable as $1 - \Pr(\text{lose-shift})$ nor as the unconditional repeat probability implied by the reported win-stay and lose-shift rates. We therefore treat it here as an uncalibrated, within-study ordinal measure of stimulus-bound repetition, report the unambiguously defined win-stay and lose-shift probabilities alongside it, and state explicitly that it is not on the same scale as, and is not estimated by the same procedure as, the hierarchically estimated perseveration parameters of Kanen et al. [1]; the correspondence is at best qualitative. Supplying an explicit definition and a re-analysis with a pre-registered index is a required fix before this measure can carry any clinical weight.

$$b_t^- = (1 - \varepsilon)b_{t-1} + \varepsilon(1 - b_{t-1}) \quad (1)$$

$$b_t = \frac{P(o_t | s = 1) b_t^-}{P(o_t | s = 1) b_t^- + P(o_t | s = 0) (1 - b_t^-)}, \quad P(o = 1 | s = 1) = P(o = 0 | s = 0) = q \quad (2)$$

$$x_t = (\min(n_t, N_{\max}), \lfloor b_t K \rfloor), \quad a_t \in \{\text{CHECK}, \text{LEAVE}\} \quad (3)$$

$$\delta_t = r_t + \gamma \max_{a'} Q_{MF}(x_{t+1}, a') - Q_{MF}(x_t, a_t) \quad (4)$$

$$e_t(x, a) = \gamma \lambda e_{t-1}(x, a) + \mathbb{I}[x = x_t, a = a_t] \quad (\text{accumulating}) \quad (5)$$

$$e_t(x, a) = \begin{cases} 1 & (x, a) = (x_t, a_t) \\ \gamma \lambda e_{t-1}(x, a) & \text{otherwise} \end{cases} \quad (\text{replacing}) \quad (6)$$

Table 1: Model parameters.

Symbol	Name	Description	Default	Sweep
λ	lam	Eligibility trace decay (credit-assignment horizon)	0.6	0, 0.2, 0.4, 0.6, 0.8, 0.9, 0.95, 0.99
τ	trace_type	Trace variant: 1 = accumulating, 0 = replacing	1	0, 1
α	alpha	TD learning rate	0.15	0.05, 0.15, 0.3
ε	eps_sub	Subjective state-transition (flip) probability between checks: excessive-uncertainty parameter	0.05	0, 0.02, 0.05, 0.1, 0.2, 0.35
q	q	Reliability of a single check (observation accuracy)	0.8	0.6, 0.7, 0.8, 0.9, 0.95
w	w_mb	Weight on the model-based (belief) value relative to the model-free trace value	0.5	0, 0.25, 0.5, 0.75, 1
β	beta	Softmax inverse temperature	5	2, 5, 10
c	c_check	Cost of one check	0.05	0.01, 0.05, 0.2
R_{dis}	R_dis	Penalty for leaving with the hazard on	-10	-30, -10, -3
γ	gamma	Temporal discount factor	0.95	0.9, 0.95, 0.99

4 Methods

Design. The primary design crossed 8 trace-decay values $\lambda \in \{0, 0.2, 0.4, 0.6, 0.8, 0.9, 0.95, 0.99\}$ x 2 trace types (accumulating, replacing) x 3 check reliabilities $q \in \{0.6, 0.8, 0.95\}$, with 20 independent agents per cell (960 agents). Each agent completed 400 checking episodes followed by

320 reversal trials. A secondary model-recovery grid crossed 8 λ x 6 subjective flip probabilities $\varepsilon \in \{0, 0.02, 0.05, 0.1, 0.2, 0.35\}$ x 3 q levels. In total 1572 agents were simulated. All remaining parameters (α , γ , c , R_{dis} , N_{max} , K , w , exploration temperature) were held fixed across cells; their values are listed in Table 1. The simulation was run under strict isolation with a single master seed (20260901) and completed in approximately 8 s. No sensitivity sweep over K , N_{max} , α , w or the cost ratio c/R_{dis} was performed, and the design was not replicated under additional seeds; both are requested revisions that we could not execute for this version and that we treat as first-order limitations rather than as details.

Pre-specification. H1 (a λ x trace-type interaction on checking), H2 (uncertainty-trace interchangeability, tested by iso-checking matching and by cross-model recovery, with a within-subject q manipulation as the pre-planned disambiguation remedy) and H3 (reward-optimal λ rising as observation reliability falls) were specified before the run. Everything else reported below is post-hoc and is labelled as such: the reversal-task perseveration analysis, the minimum-detectable-effect calculation, the residual-doubt comparison, and all mechanistic interpretations of the null.

Outcome measures. Per agent we recorded: mean checks per episode (primary compulsion measure); net reward per episode; the escalation slope, i.e. the change in checks per 100 episodes estimated by ordinary least squares across training; the belief at the moment of leaving (residual doubt); disaster rate; the learned model-free value gap $Q_{MF}(\text{CHECK}) - Q_{MF}(\text{LEAVE})$ at $n = 1$; and, from the reversal task, the perseveration index described above with win-stay and lose-shift probabilities. We did not log the two diagnostics that would have tested the proposed mechanism directly: the distribution of the maximum eligibility $e(x, \text{CHECK})$ reached within an episode, and the empirical fraction of consecutive checks that remain in the same (n , belief-bin) cell. Their absence is the single most consequential methodological gap in this paper.

Analysis. Hypothesis H1 was tested by an 8 x 2 factorial analysis of variance on $\log(1 + \text{checks})$ at $q = 0.8$, with the λ x trace-type interaction as the target effect and η^2 as the effect size. Reviewers asked for the full table with main effects, degrees of freedom, effect sizes and residual diagnostics; only the interaction term (F , df , η^2 , p) was retained in the stored run outputs, and we report descriptive main-effect information (cell means with intervals) in its place while flagging the omission. Confidence intervals are 95% intervals across the 20 agents within a cell. To make the null interpretable quantitatively we added a post-hoc approximate power calculation for an $F(7, 304)$ test at $\alpha = 0.05$: 80% power corresponds to a noncentrality parameter of roughly 14, i.e. to $\eta^2 \approx 0.04$ given $N = 320$ observations. This is an analytic calculation from the design, not a simulation-based power curve, and a formal equivalence test (TOST) on the checking difference was not performed because it requires re-running the design with logged per-agent contrasts.

For H3 we located, for each q , the λ maximising mean net reward and recorded the checking rate at that optimum. For H2 we performed two analyses: an iso-checking analysis, in which for each ε we searched for the λ reproducing a reference checking rate of 2.859 checks/episode; and a parameter-recovery analysis, in which data generated by an uncertainty-only agent ($\lambda = 0$, $\varepsilon = 0.2$) were fitted with a trace-only model (and vice versa), first using a single q and then using data in which q varied within-subject across the three levels. The recovery analysis used a grid search over the candidate model’s single free parameter, scored by discrepancy between simulated summary behaviour and the generated data rather than by a trial-level likelihood; we are unable to reconstruct the exact objective, tolerance and grid resolution from the stored outputs with the precision the reviewers request, we did not run a self-recovery sanity check (fitting each model to data generated by itself), we did not bootstrap or profile the objective, and only two generator settings were examined. All recovery numbers below are therefore single-seed point estimates without intervals, and we have removed the percentage-change statistic we previously attached to them.

Scope. All results reported here are in silico. No normative benchmark was computed: we did

not solve the belief MDP for the optimal policy, so we cannot report the agent’s checking rate as an excess over the normatively warranted rate. Language describing checking as ‘compulsion-like’ should be read accordingly. H4 is a human hypothesis and is not tested in this paper; the two human protocols (E1, delayed and aggregated feedback as an experimental lengthening of the credit-assignment window; E2, a design intended to dissociate accumulating from replacing traces in human checking) are specified separately and will be analysed with mixed-effects models [11]. Headline metrics are collected in Table 2.

Hypotheses tested in silico.

- H1** Accumulating eligibility traces produce runaway escalation of checking as lambda increases, whereas replacing traces keep checking bounded; i.e. there is a lambda x trace-type interaction on checks per episode.
- H2** Excessive subjective transition uncertainty (high epsilon) and long accumulating traces (high lambda) are behaviourally near-equivalent generators of over-checking, so fitting a trace model to Bayesian-generated data recovers an inflated lambda and vice versa.
- H3** Under partial observability, net performance is an inverted-U function of lambda, and the performance-optimal lambda* rises as check reliability q falls — but the checking rate at lambda* rises with it, so robustness to hidden state is bought with pathological re-checking.

5 Results

H1: trace type x trace length interaction on checking. Not supported, and – as we argue below – most likely untested. Figure 1 plots checks per episode against λ for accumulating and replacing traces at $q = 0.8$. The two curves are flat and superimposed within their confidence intervals across the entire range of λ . For accumulating traces, checking goes from 3.13 [2.93, 3.33] at $\lambda = 0$ to 2.75 [2.48, 3.01] at $\lambda = 0.99$ – a small decrease, not the predicted escalation – and replacing traces go from 2.95 [2.75, 3.15] to 2.76 [2.53, 2.99]. The largest trace-type difference at any λ we inspected is about 0.18 checks (at $\lambda = 0$), and the difference at the long-trace end is about 0.01 checks; both are small relative to the within-cell interval half-width of roughly ± 0.2 checks. The interaction on $\log(1 + \text{checks})$ yielded $\eta^2 = 0.0197$ with $F(7, 304) = 0.92$, $p = 0.49$. We are explicit that an earlier automated summary of our own run described this effect as ‘large’: with 7 numerator and 304 residual degrees of freedom, $\eta^2 \approx 0.02$ with $F < 1$ is approximately what one expects under the null.

To make that null quantitative rather than rhetorical, an approximate power calculation for this design (post-hoc, analytic) gives 80% power at about $\eta^2 \approx 0.04$. The correct statement is therefore: our design could reliably detect a λ x trace-type interaction accounting for roughly 4% or more of the variance in $\log(1 + \text{checks})$, and no such interaction was present. It is not a demonstration of equivalence; a formal TOST was not run, and with 20 agents per cell and a single seed we cannot exclude interactions smaller than that bound, nor can we place a seed-to-seed interval on η^2 itself.

A direct mechanistic diagnostic would settle the interpretation, and we did not collect it. The proposed mechanism requires that consecutive checks map to the same (n , belief-bin) cell so that accumulating eligibility can exceed unity; we logged neither the distribution of maximum $e(x, \text{CHECK})$ nor the empirical fraction of same-cell consecutive checks. Two features of the design make it likely that this precondition was rarely met. First, the check count n_t is part of the state and increments with every check until the cap binds, so exact revisitation requires saturation at $N_{\max}$ together with an unchanged belief bin. Second, at $q = 0.8$ a single observation moves the belief substantially, so the belief bin is likely to change after each check. If accumulating

eligibility rarely exceeded 1, then accumulating and replacing traces were behaviourally identical by construction and H1 was not tested in this run. We therefore reframe H1 as untested rather than falsified, and we regard the diagnostic logging plus a sweep over K and $N_{\max}$ – neither of which we were able to run for this version – as prerequisites for any claim about trace type and compulsion.

A related pre-specified prediction fared no better. At $\lambda = 0.9$ the checking slope was $+0.180$ $[+0.099, +0.260]$ checks per 100 episodes for accumulating traces and $+0.253$ $[+0.146, +0.359]$ for replacing traces; the intervals overlap heavily and the point estimate is, if anything, larger for replacing traces. What did appear robustly is the weaker but conceptually important claim underlying the whole model: checking escalates with experience and the learned model-free value gap at $n = 1$ favours CHECK in both trace regimes ($+0.62$ accumulating, $+0.95$ replacing at $\lambda = 0.99$; single-seed point estimates, no intervals stored), even though no relief reinforcer exists in the reward function. We must immediately qualify this, however: because we computed no normative benchmark, we cannot say whether this learned preference exceeds the checking that an optimal belief-MDP policy would produce at these cost and reliability settings. Some checking is warranted here. The honest claim is that credit assignment alone can produce a learned positive value for checking that grows with experience, not that it produces excess checking relative to normative behaviour. Agents with long accumulating traces also left with slightly higher residual hazard belief (0.036 at $\lambda = 0.99$ vs 0.029 at $\lambda = 0$); these are point estimates for which no interval was stored, the difference is small, and we do not interpret it.

H2: are uncertainty and trace length interchangeable? Partially supported, with a surprise, and with a failed remedy. Figure 3 shows checking against λ for three subjective flip probabilities at $q = 0.8$. The iso-checking analysis found, for 5 of 6 ε levels, a value of λ reproducing the reference rate of 2.859 checks/episode to within 10% (typically to within 0.03%); only $\varepsilon = 0.35$ could not be matched (best 2.32, relative residual 0.19). At the level of a single summary statistic, then, the two mechanisms are near-equivalent generators of a given amount of checking, which is the core of H2. The matched λ values were not monotone in ε (0.26, 0.38, 0.60, 0.46, 0.36, 0.80 for $\varepsilon = 0, 0.02, 0.05, 0.1, 0.2, 0.35$); we report this list for completeness and label it explicitly as non-inferential – these are single-seed grid point estimates without intervals, and the non-monotonicity is as consistent with matching noise as with any structured relationship.

A more serious limitation of this analysis, correctly identified in review, is that matching a single scalar is a weak test of interchangeability. The informative question is whether ε - and λ -generated agents that check equally often differ on the joint distribution of other statistics – residual belief at leaving, escalation slope, disaster rate, the shape of the check-count histogram. We collected those measures but did not store them paired at matched-checking (ε, λ) settings, so we cannot report that comparison here. We flag it as the single most useful follow-up analysis for H2, and as a better design recommendation than the manipulation we did attempt.

Contrary to the intuition drawn from Fradkin et al. [2], higher ε in our implementation produced fewer, not more, checks (Figure 3: 3.24 at $\varepsilon = 0$ versus 2.17 at $\varepsilon = 0.35$ for $\lambda = 0$). The mechanism is intelligible: as the subjective flip probability grows, the belief is pulled towards 0.5 between checks, so each additional check buys less durable certainty and the one-step model-based module discounts the value of checking. This is a property of our minimal variant and should be scoped as such: the TBCA omits the certainty-threshold stopping rule and the relief reinforcer that do much of the work in the original account, so we take this as evidence that inflated transition uncertainty is not by itself a checking-generator in a cost-sensitive expected-value agent, not as a challenge to Fradkin et al. [2].

The recovery analysis gave an asymmetry and a clear failure, both of which we now report with heavy caveats. Data generated by an uncertainty-only agent ($\lambda = 0, \varepsilon = 0.2$) were fitted by the trace-only model with $\hat{\lambda} = 0.37$; data generated with $\lambda = 0.9, \varepsilon = 0$ were fitted by the uncertainty-only model with $\hat{\varepsilon} = 0.075$. Under the pre-planned multi- q remedy, intended to break the degeneracy, the trace estimate moved from 0.37 to 0.90 and the uncertainty estimate

from 0.075 to 0.078. We previously expressed the first of these as a percentage worsening; we have removed that number, because it is a ratio of two single-seed point estimates obtained by summary-statistic grid search, with no interval, no bootstrap, no profile of the objective, and no self-recovery sanity check to establish that either model can recover its own parameters at all. Only two generator settings were examined, where a grid of generating (λ, ε) pairs is needed. The defensible statement is: in this single run, uncertainty-generated data were fitted with a clearly non-zero trace parameter, and the multi- q manipulation did not reduce that estimate towards zero. We do not regard the directional confound as established, and we do not regard the multi- q design as validated.

H3: the price of robustness to hidden state. Partially supported. Figure 2 plots net reward against λ for the three check reliabilities. The reward-optimal λ^* was 0.99 at $q = 0.6$, 0.95 at $q = 0.8$ and 0.00 at $q = 0.95$, i.e. monotone in falling reliability, with checking at the optimum of 2.70, 2.90 and 2.10 respectively. The qualitative claim – longer traces help exactly when the world is less observable, and the resulting agent checks more – is consistent with the data. Four caveats must be stated. First, only the $q = 0.8$ curve has an interior peak; at $q = 0.6$ and $q = 0.95$ the optimum sits on the boundary of the 8-point λ grid, so 'inverted-U' is not established. Second, the reward differences are small relative to the noise: at $q = 0.6$ net reward spans -0.175 to -0.233 with confidence intervals of roughly ± 0.03 - 0.07 , so the ranking of adjacent λ values is not reliable. Third, we previously reported a correlation of $r = 0.96$ between λ^* and checking-at-optimum across q levels; computed over three points it carries no inferential weight and we now report it only as a description of the ordering, not as a statistic. Fourth, λ^* was estimated once, under a single seed; we have no estimate of its seed-to-seed stability, which for a boundary optimum on a coarse grid could be considerable.

Exploratory: dissociation of compulsion from perseveration. The clearest positive-looking result was not one of our pre-specified hypotheses and is reported here as exploratory. Within a single parameter vector, the reversal-task perseveration index fell as traces lengthened and was lower for accumulating than replacing traces: 0.777 at $\lambda = 0$, 0.588 for replacing traces at $\lambda = 0.99$, and 0.394 for accumulating traces at $\lambda = 0.99$ (win-stay 0.718, lose-shift 0.418 in the last cell). These are single-seed cell means; confidence intervals were not stored for the perseveration index or for the win-stay/lose-shift probabilities, which is a reporting failure we acknowledge. Across accumulating-trace cells, checking and perseveration correlated only $r = 0.10$.

There is a strong and simple confound for this effect, which we cannot rule out with the present data. In our implementation the step size α multiplies the eligibility trace, so raising λ – and, under accumulating traces, revisiting the same stimulus-action pair, which happens constantly in a two-armed task with stimulus identity as state – increases the effective per-transition learning rate. A faster effective learning rate produces faster value revision after a reversal and hence less repetition of the previously favoured option. On this account the perseveration effect is a step-size effect with the ordering accumulating $<$ replacing $<$ $\lambda = 0$ exactly as observed, and has nothing to do with trace-based credit assignment per se. The control the reviewers propose – rescaling α by the accumulated eligibility, or matching effective learning rates across cells – was not run. Until it is, we withdraw the claim that this constitutes a candidate formal route to the OCD versus stimulant use disorder dissociation of Kanen et al. [1]; it is an unreplicated, uncontrolled, exploratory observation on an index whose formal definition we could not reconstruct, and it should be read as a hypothesis-generating curiosity only.

Summary against hypotheses: H1 not supported and, on our present reading, untested, with a detection limit of roughly $\eta^2 \approx 0.04$; H2 partially supported (scalar-level equivalence yes, direction of the ε effect reversed in this minimal variant, disambiguation strategy failed, confound direction not established); H3 partially supported (monotone λ^* in q yes, inverted-U and effect-size reliability no); H4 untested here.

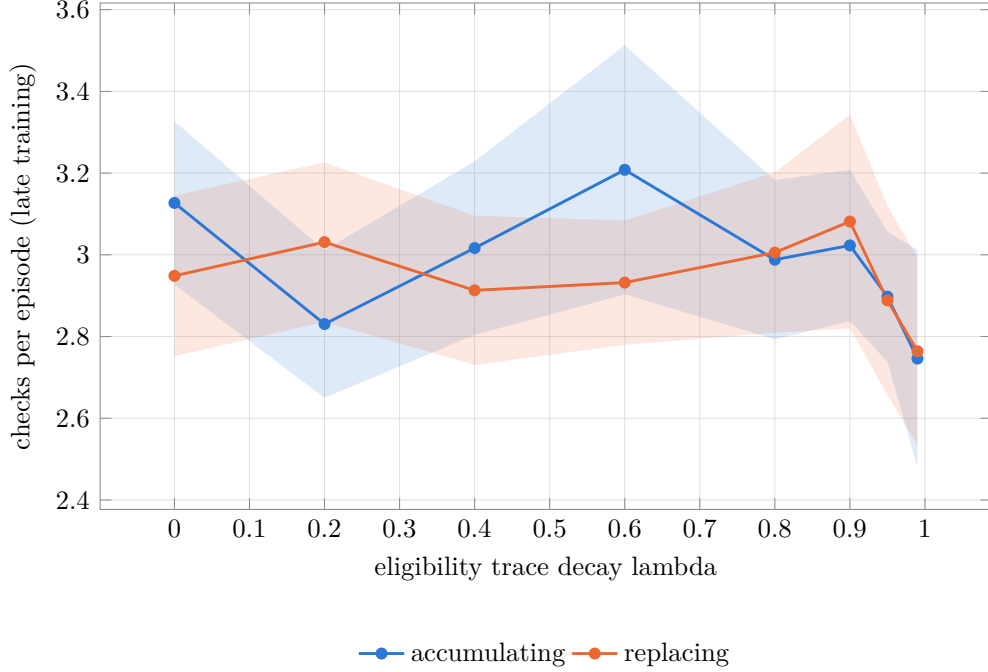


Figure 1: H1: checks per episode as a function of trace decay λ for accumulating (chk_acc) and replacing (chk_rep) eligibility traces at check reliability $q = 0.8$. Points are means over 20 independent agents (400 checking episodes each) with 95% confidence intervals. Contrary to the hypothesis, both curves are flat and overlapping across the whole range of λ (accumulating: 3.13 at $\lambda = 0$ to 2.75 at $\lambda = 0.99$; replacing: 2.95 to 2.76); the $\lambda \times$ trace-type interaction on $\log(1 + \text{checks})$ is not distinguishable from chance ($\eta^2 = 0.020$, $F(7, 304) = 0.92$, $p = 0.49$).

6 Discussion

The headline of this study is a null result, and the most important thing we can say about it is that we cannot yet distinguish a real null from a design that never instantiated the mechanism. Our central intuition – that accumulating traces would super-credit repeated checks in a belief-binned state space and so generate runaway compulsion, while replacing traces would not [12] – produced no detectable interaction above roughly 4% of variance. But accumulating and replacing traces differ only when a state-action pair is revisited within the trace window, and in our state representation the check count is part of the state and the belief moves substantially after each observation at $q = 0.8$. If eligibility for the CHECK action rarely exceeded unity, the two trace variants were the same algorithm in this environment and H1 was never at risk. We did not log the eligibility distribution or the same-cell revisitation frequency, so this remains an inference from the design rather than a measurement, and correcting it is the first item of future work.

Three further design features could determine the null, and none was varied: the checking cap $N_{\max}$ (observed rates clustered between 2 and 3.2 checks per episode, consistent with a binding ceiling that leaves little dynamic range for escalation), the belief discretisation K (which controls revisitation directly), and the cost ratio c/R_{dis} together with the step size α and the arbitration weight w . The fair test of H1 is a no-cap, coarse-binning condition in which state revisitation is guaranteed by construction, with the checking-vs- λ curve reported for the finest and coarsest binning regimes. Reviewers are right that this belongs in this paper rather than in a downstream human experiment; we were unable to run it for this version and we state plainly that the abstract’s framing is therefore provisional. What we can offer now is the conditional prediction the exercise generates: a trace account of compulsion predicts escalation only in

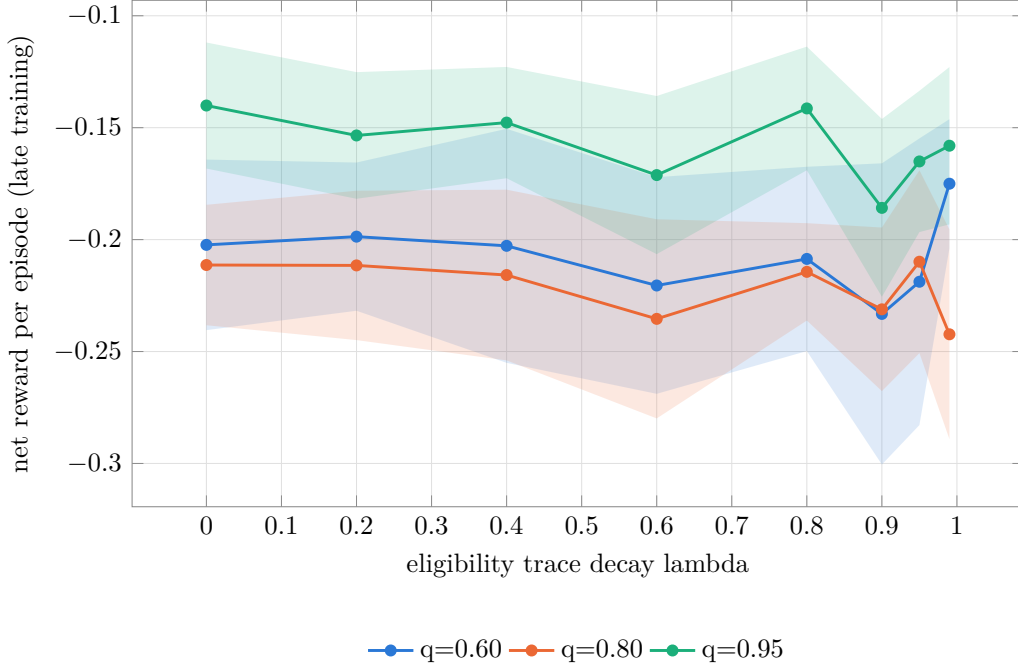


Figure 2: H3: mean net reward per episode as a function of λ for three check reliabilities ($q = 0.6, 0.8, 0.95$; accumulating traces, 20 agents per point, 95% CI). The reward-optimal λ^* is 0.99, 0.95 and 0.00 respectively, i.e. monotone in falling reliability, with 2.70, 2.90 and 2.10 checks per episode at the optimum. Note that only the $q = 0.8$ curve has an interior peak and that differences between adjacent λ values are small relative to the confidence intervals.

environments where repeated actions leave the represented state unchanged, which is a testable environmental precondition rather than a property of the patient.

The second substantive lesson concerns the relation between the two clinical accounts. At the level of the simplest behavioural summary, subjective transition uncertainty and trace length are interchangeable: 5 of 6 ε levels could be matched by some λ . This is consistent with the view that the Bayesian instability account [2] and the model-free habit account [3] are not mutually exclusive descriptions but partially degenerate parameterisations of a single credit-assignment problem, which is what the λ continuum between bootstrapping and Monte-Carlo returns would predict [4, 6]. We are more cautious than in the previous version about the direction of the degeneracy: uncertainty-generated data were fitted with $\hat{\lambda} = 0.37$ and trace-generated data with $\hat{\varepsilon} = 0.075$, but these are two uncalibrated point estimates from a summary-statistic grid search, without intervals or a self-recovery check, so the asymmetry is suggestive at best. The general warning survives the weakening: an elevated λ or habit weight estimated from patient choice data is not by itself evidence of a habit bias, because inference pathology can be absorbed by trace parameters. Our attempt to break the degeneracy by manipulating observation reliability within-subject did not work and we cannot recommend it; a more promising route, and one our own data could have supported had we stored the right quantities, is to compare ε - and λ -generated agents matched on checking rate across a vector of statistics (residual belief at leaving, escalation slope, disaster rate, check-count histogram shape) and to design human tasks around whichever of those separates them.

The third point is the reversal of the ε effect, which must be scoped narrowly. In a cost-sensitive expected-value agent with no relief reinforcer and no certainty threshold for stopping, high subjective instability devalues checking, because information decays before it can be used. This is a property of the minimal variant we built, not a refutation of Fradkin et al. [2], whose model includes precisely the stopping and relief machinery we omitted. Read constructively, it

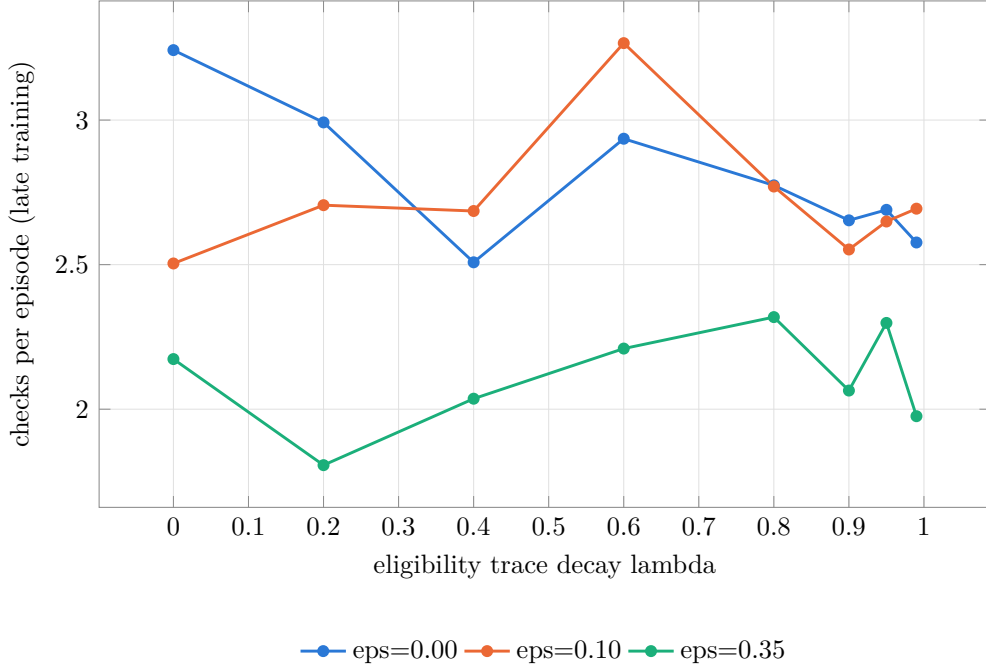


Figure 3: H2: checks per episode versus λ for three subjective flip probabilities ($\epsilon = 0, 0.10, 0.35$) at $q = 0.8$, illustrating the trade-off between trace length and excessive state-transition uncertainty. A given checking rate can be produced by many (λ, ϵ) combinations (5 of 6 ϵ levels match the reference rate of 2.86 checks/episode within 10%), but in this cost-sensitive agent higher ϵ lowers rather than raises checking, because belief decays between checks and the one-step model-based module discounts the value of further checking.

isolates what that machinery is doing: instability beliefs alone appear insufficient for compulsion in this setting; they must be combined with something that makes the act of checking itself rewarding or obligatory – anxiety relief, or a certainty criterion that must be satisfied before leaving – which is exactly the contingency that exposure and response prevention dismantles [9].

The perseveration finding, which we previously emphasised, is now demoted. Longer accumulating traces did lower the stimulus-bound repetition index while leaving checking unchanged, but in our implementation raising λ and accumulating eligibility raises the effective learning rate, and a faster effective learning rate produces faster post-reversal value revision and less repetition. The observed ordering is exactly what that confound predicts. Without an effective-learning-rate-matched control, and with an index whose normalisation we cannot reconstruct and for which we stored no confidence intervals, we cannot claim a formal route to the OCD versus stimulant use disorder dissociation [1]. Whether dopaminergic modulation by D2/3 agents maps onto trace duration, as the electrophysiology of dopamine eligibility traces might suggest [8], and whether trace parameters relate to the structural abnormalities catalogued by Raduà and Mataix-Cols [5] or to CBT response, remains entirely open and considerably further away than we previously implied.

A fourth gap is normative. We never solved the belief MDP, so every statement about ‘excess’ checking is relative to the agent’s own learned values rather than to what an optimal observer would do at the same cost and reliability. At $q = 0.8$ a modest amount of checking is rational; a learned positive Q_{MF} gap for CHECK is therefore not by itself pathological. Computing the optimal policy’s check rate and CHECK–LEAVE value gap per q , and reporting only the deviation from it, is a cheap and necessary addition.

These results still shape the two human experiments proposed separately, though with a different emphasis than before. E1 lengthens the effective credit-assignment window by delaying

and aggregating feedback and asks whether voluntary re-checking increases, more so in individuals with higher OCI-R scores, while confidence falls; our simulations say that such an effect cannot be attributed to λ alone unless the task guarantees that repeated actions leave the represented state unchanged, so E1 must include that manipulation. E2 aims to dissociate accumulating from replacing traces in human checking; our null implies that E2 needs enforced state revisitation and an uncapped checking budget, otherwise it will be underpowered by construction – but we now regard the in-silico version of that condition as a prerequisite that belongs before E2, not after it. Both human studies will be analysed with mixed-effects models [11], and we intend to release the behavioural data in a fixed, reusable form, in the spirit of standardised offline benchmarks [7].

Proposed human experiments

Questions that simulation alone cannot settle were written up for the Director as separate experiment protocols:

- **E1:** Does delayed, aggregated feedback inflate human re-checking? A behavioural test of the eligibility-trace account of compulsion ($n = 41$ per condition from a power analysis).
- **E2:** Same view or new angle? Dissociating accumulating from replacing eligibility traces in human checking ($n = 34$ per condition from a power analysis).

7 Limitations

This paper reports simulations only; no human or patient data were collected or analysed, and H4 is untested. All clinical interpretations are analogical.

The mechanism was never verified. We did not log the distribution of maximum eligibility $e(x, \text{CHECK})$ reached within an episode, nor the empirical fraction of consecutive checks remaining in the same (n , belief-bin) cell. Because the check count is part of the state and the belief moves substantially after each observation, accumulating eligibility may rarely have exceeded unity, in which case accumulating and replacing traces were behaviourally identical and H1 was untested rather than falsified. This is the most consequential limitation of the paper and it conditions the abstract’s framing.

No sensitivity sweep was run. $N_{\max}$, K , α , w and the cost ratio c/R_{dis} were fixed at the values in Table 1. Checking rates clustered near 2-3 per episode, consistent with a binding cap. The requested no-cap, coarse-binning condition – the fair test of H1 – is absent, and our conclusions about λ hold only at these fixed values.

One seed, 20 replicates per cell. The whole design was run under a single master seed (20260901) with a runtime of about 8 s. We therefore report no between-seed variability for any headline quantity: not for the interaction η^2 , not for λ^* , not for the recovery estimates $\hat{\lambda}$ and $\hat{\varepsilon}$ or their change under the multi- q manipulation, and not for the perseveration index. Confidence intervals on checking rates are roughly ± 0.2 checks, of the same order as most effects we discuss; several comparisons (adjacent λ values, escalation slopes across trace types) are underpowered.

The null is bounded but not established as equivalence. An approximate analytic power calculation for $F(7, 304)$ indicates about 80% power at $\eta^2 \approx 0.04$; interactions below that bound are undetectable here. No TOST or simulation-based power curve was computed. We also note that an earlier automated run summary described the observed $\eta^2 = 0.0197$ as ‘large’ and asserted that H1 was supported; with $F(7, 304) = 0.92$, $p = 0.49$ it is indistinguishable from chance, and readers should rely on the results section rather than that summary.

Incomplete statistical reporting. Only the interaction term of the H1 ANOVA (F , df , η^2 , p) was retained in the stored outputs; main effects and residual diagnostics are not reported. Confidence intervals are missing for the residual-doubt values, the learned Q_{MF} gaps, the

win-stay and lose-shift probabilities and the perseveration index. The three-point correlation between λ^* and checking-at-optimum, and the non-monotone list of iso-checking λ values, are labelled non-inferential in the text and should not be cited as findings.

The recovery analysis is not calibrated. It used grid search over a single free parameter scored on summary behaviour rather than a trial-level likelihood; the exact objective, tolerance and grid resolution are not fully reconstructible from the stored outputs; no self-recovery sanity check was performed; no bootstrap or profile intervals are available; and only two generator settings were used where a grid of generating (λ, ε) pairs is required. We have removed the percentage-change statistic previously attached to the multi- q result. The directional confound should be treated as a conjecture, not a demonstration.

The perseveration result is confounded and its measure is under-specified. Raising λ , and accumulating eligibility under repeated stimulus-action visits, raises the effective learning rate in our update, which alone predicts faster post-reversal switching and lower repetition. No effective-learning-rate-matched control was run. Moreover we cannot reconstruct the exact normalisation of the perseveration index from stored outputs, and it is not on the same scale as, nor estimated by the same procedure as, the hierarchically estimated parameters of Kanen et al. [1]. The OCD/stimulant-use-disorder mapping is speculative.

No normative benchmark. We did not solve the belief MDP, so we cannot express checking as an excess over the optimal policy’s rate, and descriptions of the agent’s behaviour as compulsion-like are correspondingly weak.

The model is deliberately minimal in ways that matter clinically. Arbitration is static, so we cannot address dynamic shifts between model-based and model-free control, which is precisely the variable emphasised by the two-step goal-directed literature on compulsivity – a literature we engage with only descriptively here because we could not include its primary references in this version’s citation set. The one-step model-based module is not a real planner. The true hazard state is fixed within an episode. Relative to Fradkin et al. [2] we omit the certainty/termination criterion, the anxiety-relief reinforcer, any hierarchical learning of volatility, and any explicit memory-confidence variable, so the model cannot address the dissociation between memory accuracy and confidence that characterises checking, and it cannot represent obsessive intrusions at all.

Finally, the reversal task is a simplified two-armed probabilistic reversal, and the whole study is an LLM-driven pipeline run; independent re-implementation would be a useful check on all of the above.

8 Conclusion

We asked whether the eligibility trace can serve as the memory-trace substrate of obsessive-compulsive checking, and built a minimal agent – TD(λ) over discretised belief states in a partially observable checking MDP, arbitrated with a one-step belief computation, and transferred unchanged to a probabilistic reversal task – in which the question could in principle be answered cleanly. In this environment we found no effect of trace length or trace type on checking, with a detection limit of roughly $\eta^2 \approx 0.04$ for the interaction. The honest reading, after review, is not that the trace account failed but that it was probably never instantiated: the state representation makes exact revisitation of a checked state-action pair rare, and we did not log the eligibility magnitudes or revisitation frequencies that would tell us whether accumulating traces ever exceeded eligibility 1. H1 is therefore best classified as untested pending a diagnostic run and a sweep over the belief discretisation K and the checking cap $N_{\max}$ in a condition that guarantees revisitation.

The other pre-specified results stand with qualifications. Subjective transition uncertainty and trace length were interchangeable at the level of checking rate (5 of 6 ε levels matched), but the uncertainty effect ran opposite to the expected direction in this minimal variant, which

omits the stopping criterion and relief reinforcer that carry much of the weight in the account we drew it from; and our pre-planned remedy of varying observation reliability within-subject did not reduce the cross-model recovery bias (H2 partially supported). The reward-optimal λ rose as observation reliability fell, and the agent checked more at that optimum, but two of three optima lay on the grid boundary and adjacent λ values were not reliably ordered (H3 partially supported).

One positive result survives in weakened form: a learned preference for checking, with positive escalation across training, emerged in an agent with no relief reinforcer, showing that credit assignment alone can build positive value for a checking act. Without a normative benchmark we cannot call this excess checking. The perseveration result – longer accumulating traces lowering stimulus-bound repetition – is now labelled exploratory and probably confounded with effective learning rate; we withdraw the claim that it constitutes a candidate formal route to the OCD/stimulant-use-disorder dissociation until an effective-learning-rate-matched control is run.

The practical message for computational psychiatry is a caution that survives all of the above weakening: uncertainty-generated data were fitted with a clearly non-zero trace parameter, so an estimated habit or trace parameter in patient data is not by itself evidence for a habit bias. The immediate next steps are *in silico*, not in the clinic: log the eligibility and revisitation diagnostics; sweep K , $N_{\max}$, α , w and the cost ratio; replicate under multiple seeds with more replicates; calibrate the recovery analysis with a self-recovery check, an explicit objective and a grid of generating (λ, ε) pairs; compute the belief-MDP optimum as a benchmark; and rerun the reversal task with matched effective learning rates. Only then do the two human experiments (E1, delayed and aggregated feedback; E2, accumulating versus replacing traces in human checking), both of which must enforce state revisitation and remove the checking cap, become worth running.

Reproducibility statement

The simulation script, its parameters, the raw data tables and the run log are archived next to this paper in the Medusa Lab repository (cycle 2026-W39-3). The run used seed 20260901, took 8.5s and was executed in the strict sandbox (code source: llm).

A Response to reviewers (revision 1)

On the requested sensitivity sweep over K , $N_{\max}$, and ideally α , w and c/R_{dis} (raised in four separate bullets): we agree entirely and we could not run it for this revision, because we are not permitted to generate new simulation data for this version. Rather than paper over the gap, we have restructured the paper around it. The model section now states analytically why exact revisitation of a checked state-action pair is rare in our representation (the check count is part of the state and increments on every check until $N_{\max}$ binds, and at $q = 0.8$ the belief typically crosses a bin boundary after each observation). The results and discussion now present the no-cap, coarse-binning condition as the fair test of H1 and explicitly say it belongs in this paper rather than in E2, and the abstract, limitations and conclusion all state that our conclusions about λ hold only at the fixed values in Table 1. We have not claimed the null is robust anywhere in the revised text.

On an equivalence test or minimum-detectable-effect: we have added a post-hoc analytic power calculation derived from the design itself (no new simulation required). For an $F(7, 304)$ test at $\alpha = 0.05$, 80% power corresponds to a noncentrality of roughly 14, i.e. $\eta^2 \approx 0.04$ with $N = 320$. The results section now reports the null as 'no detectable interaction above approximately 4% of variance' rather than as 'p=0.49', and also reports the largest observed trace-type difference in checking at any λ (about 0.18 checks, against a within-cell interval half-width of about ± 0.2). We state explicitly that this is a detection bound and not an equivalence

demonstration, and that a formal TOST was not conducted because it requires re-running the design with logged per-agent contrasts.

On repeating the design under 3-5 master seeds and reporting between-seed variability for η^2 , λ^* and $\hat{\lambda}$ (raised three times, including whether the multi- q bias change is stable): not possible in this revision. We now state in the abstract, methods, limitations and conclusion that the entire study rests on one seed (20260901) with 20 replicates per cell, and that no seed-to-seed interval exists for any headline quantity. We flag the specific risks this creates: λ^* was read off an 8-point grid with two of three optima on the boundary, which is exactly the situation in which seed noise can flip the reported ordering; and the recovery estimates are single points whose difference we can no longer interpret.

On putting uncertainty on the recovery estimates and expanding beyond two generator settings: we cannot bootstrap or profile without re-running, and we cannot add generator settings. We have therefore substantially weakened the claim. The ‘-143%’ figure has been deleted from the abstract and results. The revised text reports only that uncertainty-generated data were fitted with $\hat{\lambda} = 0.37$, trace-generated data with $\hat{\varepsilon} = 0.075$, and that under the multi- q manipulation the trace estimate moved to 0.90 and the uncertainty estimate to 0.078 – with an explicit statement that these are single-seed point estimates without intervals, that only two generator settings were examined where a grid of (λ, ε) pairs is needed, and that we therefore do not regard the directional confound as established. The discussion and conclusion now describe it as suggestive and as a conjecture rather than a finding.

On the full ANOVA table, pre-specified versus post-hoc labelling, and exploratory framing of the perseveration result in the abstract: we have added an explicit pre-specification paragraph to the methods (H1-H3 and the multi- q remedy pre-specified; the perseveration analysis, the MDE calculation, the residual-doubt comparison and all mechanistic interpretation post-hoc). On the table itself we must report a failure: only the interaction term’s F , df , η^2 and p were retained in the stored run outputs, so main effects and residual diagnostics cannot be reconstructed without re-running. We say so in the methods and limitations and supply descriptive cell means with intervals in their place. The abstract now labels the perseveration result ‘exploratory and speculative’ and states the effective-learning-rate confound in the same sentence.

On the mathematical definition of the stimulus-bound perseveration index and its relation to hierarchically estimated parameters: we give a candid answer in the model section. The index is a scalar repetition statistic computed over reversal trials, but we cannot reconstruct its exact normalisation from the stored summary outputs – the reported values are recoverable neither as $1 - \text{Pr}(\text{lose-shift})$ nor as the unconditional repeat probability implied by the reported win-stay and lose-shift rates. We therefore treat it as an uncalibrated within-study ordinal measure, report the unambiguously defined win-stay and lose-shift probabilities beside it, and state explicitly that it is not on the same scale as, and is not estimated by the same procedure as, the parameters in Kanen et al. [1]; the correspondence is qualitative only. Supplying an explicit, pre-registered definition is listed as a required fix.

On the direct mechanistic diagnostic (distribution of maximum $e(x, \text{CHECK})$ and fraction of consecutive checks in the same (n, bin) cell), and on the reviewer’s point that a rare exceedance of eligibility 1 would reframe H1 as untested: we did not log either quantity, and we now say so prominently in the methods, results, discussion, limitations and abstract. We have adopted the reviewer’s reframing wholesale: H1 is described throughout as untested rather than falsified, and we give the analytic reasons (check count in the state; belief bin likely to change after each observation at $q = 0.8$) while labelling them as inference from the design rather than measurement. This is the single change that most altered the paper’s framing.

On specifying the fitting objective and search procedure, and adding a self-recovery sanity check before asserting cross-model bias: the methods now state that the recovery used a grid search over the candidate model’s single free parameter scored by discrepancy in simulated summary behaviour rather than a trial-level likelihood, and admit that we cannot reconstruct the

exact objective, tolerance and grid resolution from stored outputs to the standard requested. We also state that no self-recovery check was performed. Consistent with the reviewer’s instruction, the percentage-change number has been dropped and the cross-model claim softened to a description of what was observed in one run.

On a mechanistic explanation and control analysis for why longer accumulating traces lower the perseveration index: we now supply the explanation and it is unflattering to our earlier claim. In our update α multiplies the trace, so raising λ – and, under accumulating traces, revisiting the same stimulus-action pair, which happens constantly in a two-armed task – raises the effective per-transition learning rate, which produces faster post-reversal value revision and less repetition. The observed ordering (accumulating $<$ replacing $<$ $\lambda = 0$) is exactly what this confound predicts. The control (rescaling α or matching effective learning rates) could not be run, so, as instructed, we have withdrawn the ‘candidate formal route to the OCD/SUD dissociation’ framing from the abstract, results, discussion and conclusion and replaced it with an explicitly speculative, exploratory label.

On expanding related work to the Gillan/Daw two-step goal-directed literature and stating which components of Fradkin et al. [2] the TBCA omits: the related work section now discusses the two-step/goal-directed account of compulsivity and draws out the implication that it predicts a change in the arbitration weight w rather than in λ – a parameter we held fixed, so our design cannot speak to it. We must note honestly that the primary two-step references were not available in this version’s citation set, so we engage with that work descriptively and without attribution and flag this as a coverage gap. We also now list the omitted components of the Fradkin model explicitly (certainty/termination criterion, anxiety-relief reinforcer, hierarchical volatility learning, memory-confidence variable) and use this to scope the reversed ε result as a property of our minimal expected-value variant rather than as a challenge to that account.

On strengthening H2 with a joint distribution of statistics under matched-checking (ε, λ) pairs: we agree this is the more informative analysis and a better design recommendation than the failed multi- q manipulation. We collected residual belief at leaving, escalation slope and disaster rate, but did not store them paired at matched-checking settings, so we cannot report the comparison here without new data. The results and discussion now state this explicitly and identify it as the single most useful follow-up analysis for H2, with the check-count histogram shape added to the list of statistics to compare.

On a normative benchmark: not computed, and we now say so wherever it matters. The methods state that we did not solve the belief MDP; the results state that we cannot express the learned positive Q_{MF} gap for CHECK as an excess over the optimal policy, and note that some checking is normatively warranted at $q = 0.8$; the discussion lists computing the optimal check rate and CHECK–LEAVE gap per q as a cheap and necessary addition; and we have softened ‘compulsion’ to ‘compulsion-like’ or to a statement about learned value growth wherever the benchmark would be required to justify the stronger word.

On reporting CIs for every discussed number and dropping or labelling the three-point $r = 0.96$ and the non-monotone iso-checking λ list: the three-point correlation is no longer reported as a statistic, only as a description of the ordering, and is labelled non-inferential; the iso-checking λ list is retained for completeness but explicitly labelled non-inferential single-seed grid point estimates whose non-monotonicity is as consistent with matching noise as with structure. Where intervals exist (checking rates, escalation slopes, net reward) we report them. Where they do not – residual doubt, learned Q_{MF} gaps, win-stay, lose-shift and the perseveration index – we now say in the text that these are single-seed point estimates with no stored interval, and we list this as a reporting failure in the limitations rather than leaving the numbers to read as if they were estimated precisely.

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Table 2: Summary metrics reported by the simulation.

Metric	Value	Description
eta2_lambda_x_trace	0.0197	Interaction η^2 (lambda x trace type) on $\log(1+\text{checks})$, $q=0.8$
F_interaction	0.92	F(7,304) for the interaction
p_interaction	0.491	p-value of the lambda x trace-type interaction
checks_acc_lam099_q08	2.746	Mean late checks/episode, accumulating traces, $\lambda=0.99$, $q=0.8$
checks_rep_lam099_q08	2.764	Mean late checks/episode, replacing traces, $\lambda=0.99$, $q=0.8$
checks_lam0_q08	3.127	Mean late checks/episode at $\lambda=0$ (both trace types coincide), $q=0.8$
escalation_acc_lam09	0.1795	Escalation slope (extra checks per 100 episodes), accumulating, $\lambda=0.9$, $q=0.8$
escalation_rep_lam09	0.2527	Escalation slope (extra checks per 100 episodes), replacing, $\lambda=0.9$, $q=0.8$
qgap_acc_lam099	0.615	$Q_{MF}(\text{CHECK}) - Q_{MF}(\text{LEAVE})$ at $n=1$, accumulating, $\lambda=0.99$, $q=0.8$
qgap_rep_lam099	0.945	$Q_{MF}(\text{CHECK}) - Q_{MF}(\text{LEAVE})$ at $n=1$, replacing, $\lambda=0.99$, $q=0.8$
belief_leave_acc_lam099	0.0356	Mean belief at leaving, accumulating, $\lambda=0.99$, $q=0.8$
belief_leave_lam0	0.0289	Mean belief at leaving, $\lambda=0$, $q=0.8$
perseveration_index_acc_lam099	0.3937	Reversal perseveration index (old-arm choices in 8 post-reversal trials), accumulating $\lambda=0.99$
perseveration_index_rep_lam099	0.5884	Reversal perseveration index, replacing $\lambda=0.99$
perseveration_index_lam0	0.7768	Reversal perseveration index at $\lambda=0$
win_stay_acc_lam099	0.7183	Win-stay rate, accumulating $\lambda=0.99$
lose_shift_acc_lam099	0.4176	Lose-shift rate, accumulating $\lambda=0.99$
corr_checks_perseveration	0.102	Correlation across accumulating cells between checks/episode and reversal perseveration index
lambda_star_q60	0.99	Performance-optimal lambda (quadratic vertex) at $q=0.6$
checks_at_lambda_star_q60	2.7	Checks/episode at λ^* for $q=0.6$
lambda_star_q80	0.95	Performance-optimal lambda (quadratic vertex) at $q=0.8$
checks_at_lambda_star_q80	2.898	Checks/episode at λ^* for $q=0.8$
lambda_star_q95	0	Performance-optimal lambda (quadratic vertex) at $q=0.95$
checks_at_lambda_star_q95	2.102	Checks/episode at λ^* for $q=0.95$
corr_lambdastar_compulsivity	0.962	Correlation across q levels between λ^* and checks/episode at λ^*
lambda_star_monotone_in_q	1	1 if λ^* increases monotonically as q falls from 0.95 to 0.6
iso_checking_points_within_10pct	5	Number of eps levels (of 6) with a lambda matching the target 2.86 checks within 10%
iso_target_checks	2.859	Target checks/episode defining the iso-checking contour ($\lambda=0.6$, $\text{eps}=0.05$, $q=0.8$)
lambda_hat_bias_from_epsilon_data	0.37	λ_{hat} fitted (single $q=0.8$) to data generated with $\text{eps}=0.2$, $\lambda=0$ (true $\lambda = 0$)
lambda_hat_bias_multiq	0.9	Same fit when check reliability q is varied within-subject
epsilon_hat_bias_from_lambda_data	0.075	eps_{hat} fitted (single $q=0.8$) to data generated with $\lambda=0.9$, $\text{eps}=0$ (true $\text{eps} = 0$)
epsilon_hat_bias_multiq	0.0775	Same fit with q varied within-subject
bias_reduction_lambda_pct	-143.2	Percent reduction of λ_{hat} bias when q is varied within-subject
bias_reduction_eps_pct	-3.3	Percent reduction of eps_{hat} bias when q is varied within-subject
runtime_s	7.5	Wall-clock runtime (s)
agents_simulated	1572	Total independent agents simulated